



Unit Code: PRT661

Data Science Practice

Assessment 1

Fire2Air Darwin: Explainable Next-Day Smoke and PM_{2.5} Forecasting for Greater Darwin

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Group: DAN5 – Theme 2

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1. Project Overview and Objectives

Fire2Air Darwin system provides interpretable predictions of the PM_{2.5} concentration the following day based on smoke emissions in Darwin, a region affected by bushfire smoke pollution, which has been related to mortality and admissions to hospitals (Jegasothy, et al., 2023). It combines hourly NT EPA Air Quality Network measurements with NASA FIRMS Suomi NPP (NASA FIRMS, n.d.). The study focuses on Palmerston, Winnellie and Stokes Hill between 2018 and 2024. Objectives will be to see if tomorrow's average 24-hour PM_{2.5} level is higher than 25 µg/m³ (National Environment Protection Council, 2021) (classification); predict average PM_{2.5} level (regression); estimate number of hours when it will be higher (count prediction); and identify the most influential pollution, meteorological and fire predictors by means of SHAP; based on PM_{2.5} modeling studies (Lundberg & Lee, 2017; Wu, Lin, Shi, Ye, & Fang, 2022).

2. Theme Justification

Fire2Air Darwin fits under Theme 2 since it uses historical data on air quality, weather, and fires to predict next day PM_{2.5}. The project covers all essential aspects of data science from data collection to modeling and visualization (Charles Darwin University, 2026). It is not just a simple prediction but includes three health-related predictions regarding the impact of bushfires (Jegasothy, et al., 2023). SHAP improves interpretability after previous studies on PM_{2.5} (Wu, Lin, Shi, Ye, & Fang, 2022).

3. Problem Statement

Greater Darwin experiences periods of elevated fine particulate matter (PM_{2.5}), particularly during vegetation fires in and around the region (Crabbe, 2012; Hanigan, Johnston, & Morgan, 2008; Jones, Furlaud, Williamson, Johnston, & Bowman, 2022; Johnston, Kavanagh, Bowman, & Scott, 2002). Although the NT EPA provides near-real-time and historical air-quality information, it mainly reports current and past conditions rather than forecasting what may happen the next day (Northern Territory Environment Protection Authority, 2026). This makes it difficult for communities and organisations to prepare in advance for poor air quality.

Fire2Air Darwin aims to address this gap by developing an explainable next-day forecasting system for Palmerston, Winnellie and Stokes Hill. It will combine recent PM_{2.5} measurements, local weather conditions and NASA FIRMS S-NPP fire activity to predict whether the next-day 24-hour mean PM_{2.5} will exceed **25 µg/m³** (National Environment Protection Council, 2021). The forecast could support earlier planning by schools, childcare services, community organisations and outdoor-event organisers. Rather than replacing NT EPA monitoring, Fire2Air Darwin extends its value by shifting from **observing current conditions to anticipating next-day air-quality risk**.

3.1. Research Questions

The project will investigate three connected questions:

1. **Exceedance risk:** Can recent air quality, weather and satellite-detected fires predict whether tomorrow's mean PM_{2.5} will exceed **25 µg/m³**?

2. **PM_{2.5} concentration:** How accurately can tomorrow's mean PM_{2.5} concentration be predicted?
3. **Exposure duration:** How accurately can the number of hours tomorrow with PM_{2.5} above **25 µg/m³** be predicted?

Together, these questions provide decision-makers with an understandable picture of **whether poor air quality may occur, how severe it may be, and how long it may last.**

4. Initial Architecture Diagram

4.1. End-to-End Pipeline

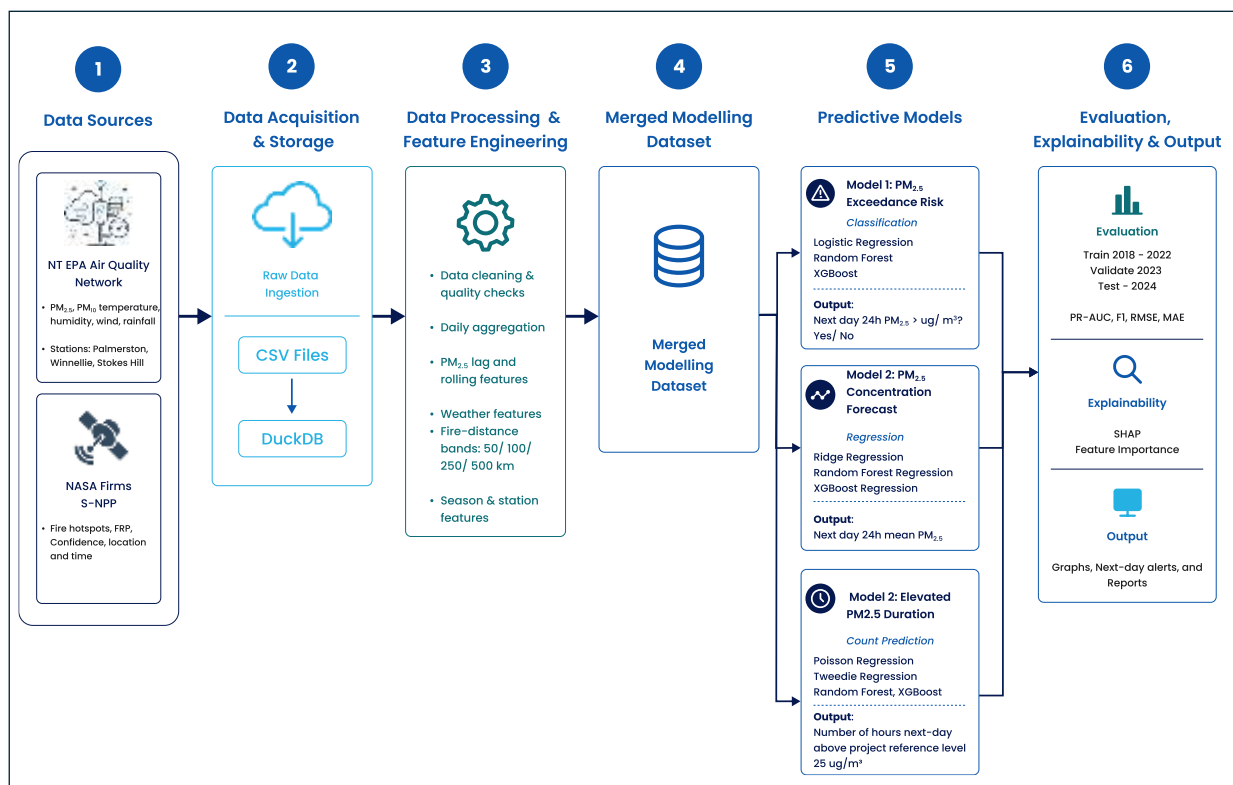


Figure 1: End-to-End Pipeline

4.2. Data Preparation and Feature Engineering

Data preparation includes quality assurance and aggregation of NT EPA hourly observations into daily records. Pollution persistence will be characterised by rolling averages and lags of PM_{2.5} and meteorological conditions with weather variables like temperature and humidity. S-NPP detections will be screened for proximity to Darwin and aggregated into daily fire and environmental features. The datasets will be merged by date to form a leakage-safe station-day modelling dataset. The missing values will be analysed for trends and appropriate imputation methods will be applied to make the pre-processing step transparent and reproducible.

4.3. Candidate Model & Model Selection

Fir2Air Darwin is evaluating several candidate models for different prediction tasks instead of selecting a single algorithm beforehand. We will compare the performance of Logistic Regression with Random Forest and XGBoost classifiers on predicting the risk of PM_{2.5} exceedance. The prediction of PM_{2.5} concentration for the next

day will be tested using Ridge Regression, Random Forest Regression and XGBoost Regression. We will consider Random Forest, XGBoost, Poisson Regression and Tweedie Regression for elevated PM 2.5 duration. The best model for each task will be chosen based on validation metrics for performance, which encourages evidence-based model selection. Then we will use SHAP values to explain how predictions are influenced by recent PM2.5, weather, and nearby fires (Gaganjot Kaur, 2018; Wei Huang, 2021).

4.4. Model Evaluation & Selection

For each prediction task, appropriate metrics will be used to evaluate its output. For Model 1 – PM_{2.5} Exceedance Risk, the primary evaluation measures are PR-AUC, recall, precision and F1-score, with ROC-AUC as a secondary measure. Class weighting will also be considered, if applicable, especially if high-PM_{2.5} days are infrequent compared to normal days.

The performance of Model 2 – PM_{2.5} Concentration Forecast will be evaluated using MAE, RMSE and R2 and candidate models will be compared to a simple persistence baseline. For Model 3 – Elevated PM_{2.5} Duration, we will assess predictions of the number of elevated-PM_{2.5} hours using MAE and RMSE and an appropriate count-model evaluation measure.

5. Workflow Plan

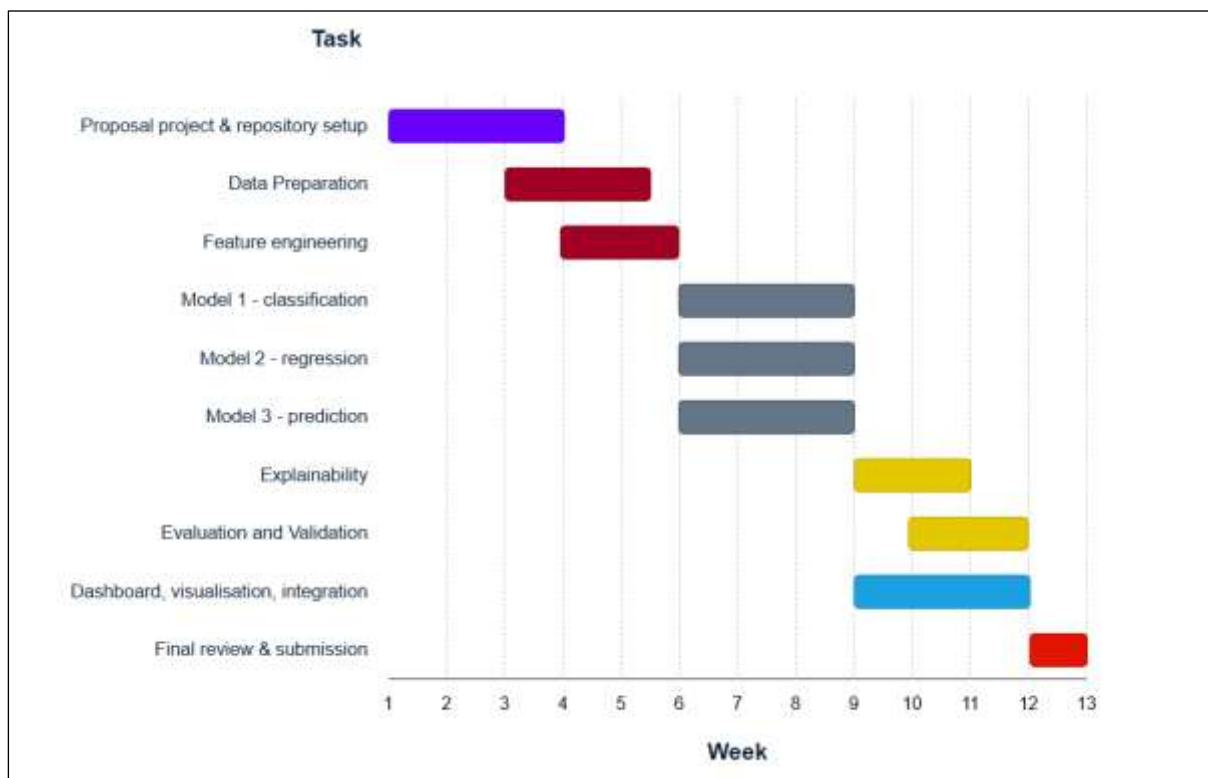


Figure 2: Workflow Plan

Figure 2 depicts the semester process starting from the repository creation, data validation and finding a leakage-free feature set in the first few weeks. Weeks 6-9: Develop three prediction tasks in parallel, each with a number of candidate algorithms to be evaluated. After the model development, model evaluation, explainability, dashboard development and visualisation will be done and visualisations will be based on validated results.

6. Task Allocation and Team Structure

Member	Role	Responsibility
Thi Pham	Data processing and feature engineering	Clean the data and prepare features for modelling.
Lihini Jinanjalie	Model 1 – Classification	Build and compare classification models and evaluate them using PR-AUC.
Esangbedo Favour	Model 2 – Regression	Build the PM _{2.5} concentration model and evaluate it using MAE and RMSE.
Navodya Piumanthi	Model 3 – Count Prediction and explainability	Develop the count model and explain model results using SHAP or permutation importance.
Dieu Yen Diep	Visualisation and documentation	Prepare the dashboard, diagrams, README and project documentation.

Table 1: Task Allocation

Each member has an assigned task, with Teams, GitHub and Jira used for communication, file sharing and progress tracking.

- **GitHub repository:** [Click here](#)
- **Jira board:** [Click here](#)

7. Initial Risk Analysis

Risk	Possible Impact	Mitigation
Missing or invalid air-quality data	May reduce model accuracy	Check data quality and handle missing or invalid values before modelling.
FIRMS fires may not affect Darwin	Could create incorrect links with PM _{2.5}	Consider fire distance, wind direction and weather conditions.
Few high-PM _{2.5} days	May make classification difficult	Use PR-AUC, recall, F1-score and class weighting where needed.
Data leakage	May give unrealistic results	Use chronological splitting and only past/current data for prediction.
Prediction errors	May reduce forecast usefulness	Compare with baseline models and report limitations.
Team delays	May affect milestones	Track tasks through Jira, GitHub and regular meetings.

Table 2: Initial Risk Analysis for Next-Day PM_{2.5} Forecasting

8. Ethics, Privacy and Security Considerations

Fire2Air Darwin uses air-quality and satellite fire data and does not involve personal or identifiable information, so privacy risk is low. Incorrect forecasts could affect decisions made by people who are sensitive to smoke exposure. The system will therefore be presented as a research and decision-support tool, not as medical advice

or an official health warning (Rudin, 2019; Ghassemi, Oakden-Rayner, & Beam, 2021). Particular attention will be given to false-negative predictions, as missing a high-PM_{2.5} day may have greater consequences.

NASA FIRMS and NT EPA data sources and usage requirements will be properly acknowledged. Raw and processed data will be stored separately, with project changes tracked through GitHub. Passwords, API keys and login details will not be uploaded and will be protected using environment variables and .gitignore.

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